



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF BIOMEDICAL ENGINEERING
B.E. Biomedical Engineering
CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2	Exempted for LE	HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3	Exempted for LE	HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
4	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
5	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
6	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
7		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
8		BS	SH2223	19MA223	Linear Algebra and Probability	-	4	0	4	CGPA	Theory
9	Exempted for LE	BS	SH3216	19PH216	Physics for Electronics Engineering	-	3	1	4	CGPA	TIP (6 Expts)
10		BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
11	Exempted for LE	BS	SH4802	NA	Chemistry for Biomedical Engineering	-	2	1	3	CGPA	TIP (6 Expts)
12		BS	EC4412	19BM412	Anatomy and Human Physiology	-	3	0	3	CGPA	Theory
13	Exempted for LE	ES	EC4301	19BM301	Electronic Circuits and Special Devices	-	2	1	3	CGPA	TIP (6 Expts)
14	Exempted for LE	ES	EC4801	NA	Analog and Digital Integrated Circuits	EC4301	3	0	3	CGPA	Theory
15	Exempted for LE	ES	EC4802	NA	Integrated Circuits laboratory	EC4301	0	1	1	CGPA	Practical (6 Expts)
16		ES	EC4803	NA	Basics of Signals and Systems	SH2201	3	0	3	CGPA	Theory
17	Exempted for LE	ES	EC2301	19EE301	Electric Circuits and Networks	19MA202	3	1	4	CGPA	TIP (6 Expts)
18		ES	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
19		ES	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
20		ES	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
21		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)

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22		ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
23		ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)
24		PC	EC4401	19BM401	Biochemistry	-	3	1	4	CGPA	TIP (6 Expts)
25		PC	EC4804	NA	Biomedical Instrumentation	EC4801, EC4802	3	1	4	CGPA	TIP (6 Expts)
26		PC	EC4805	NA	Clinical Instrumentation	-	3	1	4	CGPA	TIP (6 Expts)
27		PC	EC4806	NA	Medical Imaging System I	-	3	0	3	CGPA	Theory
28		PC	EC4807	NA	Medical Imaging System II	-	3	0	3	CGPA	Theory
29		PC	EC4404	19BM404	Pathology and Microbiology	-	3	1	4	CGPA	TIP (6 Expts)
30		PC	EC4808	NA	Clinical Laboratory Equipment and Techniques	EC4805	2	1	3	CGPA	TIP (6 Expts)
31		PC	EC4809	NA	Therapeutic Equipments I	-	3	1	4	CGPA	TIP (6 Expts)
32		PC	EC4810	NA	Therapeutic Equipments II	-	3	1	4	CGPA	TIP (6 Expts)
33		PC	EC4811	NA	Lasers in Healthcare Applications	-	3	0	3	CGPA	Theory
34		PC	EC4410	19BM410	Bio Materials and Artificial Organs	EC4406/19B M406	3	0	3	CGPA	Theory
35		PC	EC4408	19BM408	Biomedical Signal Processing	EC4803/19E C402	3	1	4	CGPA	TIP (6 Expts)
36		PC	EC4530	19MD406	Medical Image Processing	19BM408/EC 4408	3	1	4	CGPA	TIP (6 Expts)
37		PC	EC4820	19BM530	Embedded Systems and IoMT	EC4801, EC4802	3	1	4	CGPA	TIP (6 Expts)
38		PC	EC4402	19MD402	Bio Control Systems	-	3	0	3	CGPA	Theory
39		PC	EC4406	19BM406	Biomechanics	-	3	0	3	CGPA	Theory
40	Exempted for LE	PC	EC4812	NA	Sensors and Measurements	-	2	1	3	CGPA	TIP (6 Expts)
41		PE(Honours 1)	EC4518	19BM518	Foundation Skills in Integrated Product Development	-	3	0	3	CGPA	Theory
42		PE(Honours 1)	EC4519	19BM519	Medical Device Design	-	3	0	3	CGPA	Theory
43		PE(Honours 1)	EC4520	19BM520	Medical Device Regulations	-	3	0	3	CGPA	Theory
44		PE(Honours 1)	EC4521	19BM521	Patient Safety, Standards and Ethics	-	3	0	3	CGPA	Theory
45		PE(Honours 1)	EC4522	19BM522	Medical Innovation and Entrepreneurship	-	3	0	3	CGPA	Theory
46		PE(Honours 1)	EC4523	19BM523	Rapid Prototyping	-	3	0	3	CGPA	Theory
47		PE(Honours 2)	EC4524	19BM524	Clinical Engineering	-	3	0	3	CGPA	Theory

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48		PE(Honours 2)	EC4525	19BM525	Hospital Planning and Management	-	3	0	3	CGPA	Theory
49		PE(Honours 2)	EC4526	19BM526	Medical Waste Management	-	3	0	3	CGPA	Theory
50		PE(Honours 2)	EC4527	19BM527	Economics and Management for Engineers	-	3	0	3	CGPA	Theory
51		PE(Honours 2)	EC4528	19BM528	Biostatistics	-	3	0	3	CGPA	Theory
52		PE(Honours 2)	EC4529	19BM529	Forensic Science in Healthcare	-	3	0	3	CGPA	Theory
53		PE	EC4531	19BM531	Virtual Reality and Augmented Reality in Health care	-	2	1	3	CGPA	TIP (6 Expts)
54		PE	EC4413	19BM413	Virtual Instrumentation for Clinical Applications	19MD403	3	1	4	CGPA	TIP (6 Expts)
55		PE	EC4511	NA	Brain Computer Interface and its Applications	-	2	1	3	CGPA	TIP (6 Expts)
56		PE	EC4506	19BM506	Medical Nanotechnology	-	3	0	3	CGPA	Theory
57		PE	EC4532	19BM532	Medical Informatics	-	3	0	3	CGPA	Theory
58		PE	EC4507	19BM507	Robotics in Medicine	-	3	0	3	CGPA	Theory
59		PE	EC4510	19BM510	Ergonomics	-	3	0	3	CGPA	Theory
60		PE	EC4502	19BM502	Biofluids and Dynamics	19BM406	3	0	3	CGPA	Theory
61		PE	EC4814	NA	BioMEMS	-	2	1	3	CGPA	TIP (6 Expts)
62		PE	EC4815	NA	Haptics	-	3	0	3	CGPA	Theory
63		PE	EC4833	19MD507	Physiological Modeling	-	3	0	3	CGPA	Theory
64		PE	EC4504	19BM504	Rehabilitation Engineering	EC4406/19BM406	3	0	3	CGPA	Theory
65		PE	CS1509	19CS509	Graph Theory and Applications	19MA206	3	0	3	CGPA	Theory
66		PE	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
67		PE	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
68		PE	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/19CS303	3	1	4	CGPA	TIP (6 Expts)
69		PE	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
70		EEC	EC4701	19BM701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
71	Option1	EEC	EC4702	19BM702	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
72	Option1	EEC	EC4703	19BM703	Project Work Phase II	EC4702/19BM702	0	6	6	CGPA	Project
73	Option2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project

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74		EEC	EC4819	NA	Hospital Training	-	0	2	2	CGPA	Practical (10 Expts)
75		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
76		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
77		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
78		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
79		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
80		EEC	#N/A	#N/A	Placement Readiness Assessment	19EY704 / 19EY711/SH 6711	1	0	1	CGPA	Theory (Int Only)
81		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
82		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
83		OE	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
84		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
85		MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
86		MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
87		MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
88	Any 1	MC	NA	19MC801	Professional Ethics	-	0	0	0	NON - CGPA	Theory (Int Only)
89	Any 1	MC	NA	19MC802	Environmental Science	-	0	0	0	CGPA	Theory (Int Only)
90	Any 1	MC	NA	19MC803	Constitution of India	-	0	0	0	CGPA	Theory (Int Only)
91		MC	NA	19MC804	Internship/Entrepreneurship/Consultancy	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	10	0
Basic Science	23	12
Engineering Science	30	18
Professional Core	60	57
Professional Electives	18	18
Open Electives	7	7
Employability Enhancement Courses	18	18
Mandatory Courses	2	2
TOTAL	168	132